RIPHAH INTERNATIONAL UNIVERSITY, LAHORE

FINAL YEAR PROJECTS (FYP) GUIDELINES

Research-Category Projects

Operational Guidelines and Governance Policy

Document Attribute	Details
Document Title	Operational Guidelines and Governance Policy for Research-Category Final Year Projects
Target Audience	Final Year Undergraduate Students, FYP Supervisors, Evaluation Panel Members, and relevant FYP Coordinators/Committees
Academic Discipline	Computing Disciplines including Computer Science, Software Engineering, Data Science, Artificial Intelligence, Cybersecurity, and related fields
Document Version	Version 1.0
Effective Date	Fall 2026 Onwards
Issuing Authority	Riphah School of Computing & Innovation (RSCI), Riphah International University, Lahore
Document Status	Approved Operational Policy & Student Directive

Policy precedence

This document shall be read together with applicable Riphah International University regulations, FYP procedures, academic-integrity requirements, HEC policies/frameworks, research-ethics requirements, intellectual-property provisions, data-protection obligations, and applicable law. Where a higher-authority requirement conflicts with this document, the higher-authority requirement shall prevail.

Version 1.0 — Fall 2026 Onwards

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Document Control and Policy Interpretation

Version 1.0 revises the original Research-Category FYP Guidelines by retaining its research-to-artifact philosophy while adding governance provisions for research ethics, data management, academic integrity, intellectual property, version control, reproducibility, assessment, remediation, and exceptions.

Policy Term	Meaning in this document
Shall / Must	Mandatory requirement. Student compliance is expected unless an approved exception applies.
Will	Expected student action or process outcome. Where a statement is intended as a binding requirement, “shall” or “must” is preferred.
Should	Recommended practice that is normally expected where relevant and feasible.
May	Permitted or optional action; not a universal requirement.
Applicable / Where Relevant	A requirement applies when it is appropriate to the approved research problem, discipline, data, methodology, or technical artifact.

1. Purpose and Objectives of the Guidelines

The purpose of these operational guidelines is to establish clear, student-focused, academically rigorous, and actionable requirements for undergraduate students undertaking Final Year Projects (FYPs) within the Research Category across computing disciplines.

Research-category FYPs shall integrate appropriate research methodology, computing theory or domain knowledge, empirical investigation, and software or technical implementation. The project shall demonstrate a meaningful relationship between the identified research problem, the adopted methodology, the technical contribution, the evidence generated through experimentation or evaluation, and an appropriate final research artifact.

The policy shall not require every research project to follow an identical implementation pattern. Instead, students shall demonstrate the following core outcomes according to the approved nature of their project. **Students shall:**

- Formulate a clear and **researchable problem** supported by relevant scholarly literature.
- Identify a defensible **research gap, motivation, or contribution** opportunity and shall formulate appropriate research questions and objectives.
- Apply an appropriate theoretical, conceptual, mathematical, computational, or domain **framework** to guide the research.
- Use an appropriate **research methodology** and shall document the methodology sufficiently for evaluation and, where feasible, reproducibility.
- Develop and demonstrate a **technical or research artifact** appropriate to the approved project, such as a software application, functional product, algorithm implementation, computational pipeline, simulation, visualization, cybersecurity tool, experimental framework, or other suitable artifact.
- Evaluate the proposed contribution using **appropriate metrics**, baselines, experiments, or qualitative/quantitative methods justified by the research problem.
- Maintain academic and research integrity and shall appropriately **acknowledge external sources**, data, software, models, tools, and intellectual contributions.
- Comply with applicable **research-ethics**, privacy, data-governance, intellectual-property, security, and legal requirements.
- Maintain appropriate **records of source code**, datasets, experiments, configurations, and project decisions to support verification and reproducibility, subject to applicable restrictions.
- Transparently document material **use of artificial-intelligence tools** and shall remain responsible for the accuracy, integrity, originality, security, and ethical appropriateness of submitted work.

2. Classification of FYP Requirements

To ensure academic rigor without imposing unnecessary technical constraints on different computing disciplines, the requirements are classified into three categories.

2.1 Core Mandatory Requirements

The following requirements shall apply to every Research-Category FYP unless an approved exception is granted. **Students shall satisfy the core requirement of:**

- Approved research problem, objectives, and scope;
- Appropriate scholarly literature and research-gap analysis;
- Appropriate research methodology;
- Research ethics and data-governance compliance where applicable;
- Originality, academic integrity, and appropriate attribution;
- Appropriate technical/research artifact;
- Evidence-based evaluation;
- Documented supervision and milestone progress;
- Final report, demonstration, and oral defense according to the approved fyp process;
- Submission/archiving of required code, documentation, and artifacts subject to ip, privacy, licensing, and confidentiality restrictions.

2.2 Project- and Discipline-Dependent Requirements

The following requirements shall be applied where relevant to the approved project. **Students shall address:**

- Mathematical formulations, objective functions, optimization constraints, or formal models; where applicable and as approved in the project plan.
- Dataset collection, exploratory data analysis, and train/validation/test partitioning; where applicable and as approved in the project plan.
- Live model inference or deployment; where applicable and as approved in the project plan.
- Address user interfaces, APIs, dashboards, or mobile applications; where applicable and as approved in the project plan.
- State-of-the-art benchmarking; where applicable and as approved in the project plan.
- Ablation studies, statistical significance testing, robustness analysis, or computational-cost analysis; where applicable and as approved in the project plan.
- Human-subject participation, interviews, surveys, experiments, or collection of personal/sensitive data; where applicable and as approved in the project plan.
- Security testing, threat modelling, privacy assessment, or secure deployment; where applicable and as approved in the project plan.
- Specialized hardware, cloud resources, or external services. Where applicable and as approved in the project plan.

2.3 Enhancement Directives

Enhancement directives are recommended practices that may strengthen the academic or professional quality of a project but shall not automatically be treated as universal pass/fail requirements. **Students should consider:**

- Publication-oriented formatting or preparation of a manuscript; where feasible, relevant, and permitted.
- Consider public demonstration or external deployment; where feasible, relevant, and permitted.

- Open-source release of code or documentation where permitted; where feasible, relevant, and permitted.
- Additional comparative experiments beyond the approved scope; where feasible, relevant, and permitted.
- Advanced visualization, user testing, or performance optimization; where feasible, relevant, and permitted.
- Preparation of a conference/journal manuscript based on the fyp. Where feasible, relevant, and permitted.

3. AI Tool Usage Policy

This policy governs responsible use of artificial-intelligence and generative-AI tools during project research, software development, experimentation, documentation, and report preparation. **Students shall:**

- Disclose material use of AI tools in project documentation, including the tool used, purpose of use, and nature of assistance provided.
- Remain responsible for the accuracy, originality, integrity, security, and ethical appropriateness of all submitted work, including material generated or assisted by AI tools.
- Independently verify AI-generated factual claims, references, mathematical content, code, algorithms, experimental interpretations, and other substantive outputs before incorporating them.
- Ensure that AI-generated or AI-assisted code does not introduce licensing, security, privacy, or intellectual-property violations.
- Remain able to explain, defend, modify, and maintain any substantive code, methodology, analysis, or research claims submitted as part of the FYP.

Students shall not:

- Submit AI-generated content as original student work without appropriate disclosure where disclosure is required.
- Enter confidential, personally identifiable, restricted, proprietary, or otherwise protected project information into external ai services unless authorized.
- Use AI tools to fabricate data, research results, citations, participant responses, experimental evidence, or other research materials.
- Treat AI output as an authoritative scholarly source unless the output is independently verified through appropriate primary or authoritative sources.

Students may:

- Use commercial or open-source AI APIs as components of the developed system where such use is relevant to the approved research objectives and is properly documented and attributed.
- Use approved AI tools for permitted academic and software-development activities, subject to HEC, Riphah International University policies and project-specific restrictions.

Students should:

Retain representative records of significant AI-assisted work where those records are necessary to explain or verify the student's contribution; the student shall not be required to disclose confidential prompts or protected information.

4. Project Deliverables, Milestones, Change Control, and Supervisor Consultation Protocol

4.1 Standard Project Milestones

Milestone	Minimum expected evidence
M1 — Proposal & Approval	Problem, motivation, research gap, questions, objectives, scope, initial literature, proposed methodology, expected artifact, preliminary data/resource plan.
M2 — Research Design	Detailed methodology, literature synthesis, theoretical/computational framework, dataset/ethics plan where applicable, architecture/workflow, evaluation plan.
M3 — Baseline / Initial Implementation	Dataset preparation or experimental setup, baseline implementation where applicable, initial artifact, preliminary results, repository evidence.
M4 — Main Contribution	Proposed method/technical contribution, implementation, experiments, documented revisions, and supervisor-reviewed progress.
M5 — Evaluation & Analysis	Comparative evaluation, analysis of results, limitations, validity considerations, reproducibility evidence, and completed artifact.
M6 — Final Submission & Defense	Final thesis/report, final code/artifact/archive, demonstration, presentation, viva/defense, and required declarations.

4.2 Student Responsibilities

Student shall:

- Attend and prepare for scheduled supervision and milestone reviews.
- Present substantive evidence of progress rather than relying solely on verbal progress reports.
- Maintain records of major decisions, revisions, experiments, and supervisor feedback where required.
- Address reasonable supervisor feedback in subsequent revisions or shall document a justified alternative where the feedback is not adopted.
- Communicate material risks to the timeline, data availability, infrastructure, ethics, or methodology as soon as reasonably possible.

4.3 Supervisor Responsibilities

Supervisor shall:

- Provide academic guidance appropriate to the supervisor's expertise and the approved project.
- Review major research decisions, methodology, scope, progress, and significant risks at appropriate milestones.
- Advise the student when a proposed methodology, dataset, deliverable, or technical requirement appears infeasible, unethical, unsafe, or inconsistent with the approved scope.
- Provide feedback within the applicable FYP process and shall maintain appropriate evidence of milestone review where required.
- Not be treated as the author of the student's work merely because supervisory guidance was provided.

4.4 Change Control

Student shall:

- Obtain supervisor/FYP authority approval for material changes to the research problem, research questions, objectives, principal methodology, dataset/source, or final artifact.
- Document the reason, expected impact, and revised plan for material changes.

- Not use a late-stage change of scope to avoid an unresolved methodological or evaluation requirement without approval.

----------The End----------